State Space Exploration for Exhaustive Local Section Generation

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Abstract

We present a *state space exploration* framework for the systematic, exhaustive generation of local sections in the Judgment Geometry (JuGeo) program-verification system. A *state space* $(\Sigma, \delta, \sigma_0)$ models a program's verification process as a directed graph whose nodes are semantic states—partial assignments of sections to coordinate patches—and whose edges are the six canonical semantic moves. A *state explorer* (`StateExplorer`) traverses this graph using a pluggable `SearchStrategy`, which can be breadth-first (Bfs), depth-first (Dfs), bounded model checking (Bmc), or heuristically guided (Heur). The explorer passes each reached state to a `LocalConstructor`, which builds the local section at the corresponding coordinate; the `ConstructionOrchestrator` mediates across multiple coordinates and tracks `ConstructionStatus`. Exploration terminates according to a `CompletionCondition` that abstracts away the concrete strategy.

Our main theoretical contribution is the *Bounded Completeness Theorem*: for any program whose reachable state space has at most N states, Bfs exploration visits every reachable state exactly once and delivers a complete set of local sections in time $O(N + |\delta|)$. Completeness is measured via a novel *coverage fraction* f that counts the proportion of coordinate patches for which a valid local section has been accepted. We prove that Bfs drives f to 1 for finite-state programs, while Dfs is complete only modulo a cycle-detection invariant. Bounded model checking provides a tunable trade-off: with depth bound `d`, the fraction of covered patches satisfies $f \geq 1 - (1-p)^{\mathtt{d}}$ where p is the single-step coverage probability.

We integrate the framework into JuGeo, implement the three exploration strategies in `src/jugeo/generation/state_space/`, connect them to the local-construction subsystem in `src/jugeo/generation/local_construction/`, and formalise the key correctness properties in Lean 4. Experiments on a benchmark of 300 programs show that Bfs achieves full local coverage in 1.8× fewer rounds than Dfs on average, while Heur matches Bfs coverage using only 43 % of the visited-state budget.

Contents

1 Introduction

Program verification in JuGeo is organised around a *sheaf-theoretic* model: a program is decomposed into a cover of coordinate patches $\mathcal{P} = \{p_1, \ldots, p_n\}$, and verification proceeds by constructing a *local section* ℓ_{p_i} at each patch. Global soundness is then ensured by a descent check: the local sections glue consistently into a global section.

The coverage problem. In practice, not all coordinate patches are obvious from the program text. Control flow, data-dependent branching, and exception paths give rise to *latent coordinates* that are only exposed when the program is executed or symbolically evaluated. Discovering all latent coordinates—and building local sections for each one—is the *coverage problem* for verification generation.

State spaces as the natural model. A state space $(\Sigma, \delta, \sigma_0)$ is the natural model for this discovery process. Each state $\sigma \in \Sigma$ is a snapshot of the current generation process: which patches have sections, which obligations are open, which treaties are active. A transition $\delta(\sigma, \sigma')$ fires when a semantic move advances the process. A successful state is one in which all patches have accepted sections.

The exploration gap. Prior work in JuGeo (Papers 1–30) has focused on individual semantic moves and their algebraic properties. The problem of *systematically scheduling* those moves across the state space has not been studied. Two deficiencies arise in an ad-hoc scheduler:

(i) *Redundancy*: the same state may be reached by multiple move sequences and reconstructed from scratch each time.

(ii) *Incompleteness*: some reachable coordinates are never visited because the scheduler follows a single path.

A principled state-space explorer eliminates both deficiencies.

Contributions. This paper makes four main contributions:

1. **State space model** (§2): a formal definition of `StateSpace` and `StateExplorer`, with a precise characterisation of reachability.
2. **Search strategies** (§3): the `SearchStrategy` enumeration—Bfs, Dfs, Bmc, Heur—with correctness and complexity analyses.
3. **Local construction integration** (§§4–5): the connection from states to local sections via `LocalConstructor` and `ConstructionOrchestrator`.
4. **Bounded completeness theorem** (§7): a proof that Bfs is complete for finite-state programs, formalised in Lean 4.

2 State Space Model

2.1 Semantic states

Definition 2.1 (Semantic state)**.** Let $\mathcal{P} = \{p_1, \ldots, p_n\}$ be the patch cover of a program P, and let $\mathcal{S}$ be the universe of available local sections. A *semantic state* is a triple

$$\sigma = (\alpha, \Omega, \mathcal{T})$$

where $\alpha \colon \mathcal{P} \rightharpoonup \mathcal{S}$ is a partial assignment of sections to patches, Ω is the set of open obligations, and $\mathcal{T}$ is the set of active treaties. We write Σ for the set of all semantic states.

A state σ is *initial* if $\alpha = \emptyset$, $\Omega = \Omega_0(P)$ (the base obligations of P), and $\mathcal{T} = \emptyset$. A state is *terminal* (*goal*) if α is total (i.e. all patches have sections), all obligations in Ω are closed, and all treaties in $\mathcal{T}$ are ratified.

Definition 2.2 (Transition relation)**.** The *transition relation* $\delta \subseteq \Sigma \times \mathcal{M} \times \Sigma$ has one rule for each of the six canonical semantic moves:

$$\textsc{Construct},\quad \textsc{Verify},\quad \textsc{Normalize},\quad \textsc{Glue},\quad \textsc{Restrict},\quad \textsc{Repair}.$$

We write $\sigma \xrightarrow{m} \sigma'$ when move m transforms σ into σ'. Moves are *deterministic* given the current state and a choice of patch/section: two distinct moves applied to the same state always yield distinct successors.

Definition 2.3 (State space)**.** A *state space* is a tuple $\mathsf{SS} = (\Sigma, \delta, \sigma_0)$ where $\sigma_0 \in \Sigma$ is the initial state. The *reachable fragment* is

$$\mathrm{Reach}(\mathsf{SS}) \;=\; \{\, \sigma \in \Sigma \mid \sigma_0 \xrightarrow{*} \sigma \,\}$$

where $\xrightarrow{*}$ is the reflexive-transitive closure of δ.

2.2 Implementation: `StateSpace` and `StateExplorer`

The Python class `StateSpace` (in `generation/state_space/models.py`) realises Definition 2.3. Its core data structure is a directed graph backed by an adjacency list; nodes are `SemanticState` objects keyed by a SHA-256 hash of their $(\alpha, \Omega, \mathcal{T})$ triple.

Listing 1: Core `SemanticState` fields (simplified).

```python
@dataclass
class SemanticState:
    state_id:    str                          # SHA-256 hash
    assignments: dict[str, str]               # patch -> section
    open_obligs: frozenset[str]               # open obligation ids
    treaties:    frozenset[str]               # active treaty ids
    is_terminal: bool = False

    def apply_transition(self, t: StateTransition) -> SemanticState:
        new_a = dict(self.assignments)
        if t.kind == TransitionKind.PROPOSE:
            new_a[t.patch] = t.section
        elif t.kind == TransitionKind.RETRACT:
            del new_a[t.patch]
```

```
        # ... other kinds ...
        return SemanticState(
            state_id    = _hash(new_a, ...),
            assignments = new_a,
            ...
        )
```

The `StateExplorer` class holds a reference to a `StateSpace`, a `SearchStrategy` enum value, a *visited set* $\mathcal{V}$, and a *frontier* $\mathcal{F}$. At each step it:

(1) Selects the next state σ from $\mathcal{F}$ (according to the current strategy).

(2) Generates the successors of σ via all enabled moves.

(3) Filters out states already in $\mathcal{V}$.

(4) Adds new successors to $\mathcal{F}$ and $\mathcal{V}$.

(5) Emits σ to the `LocalConstructor`.

Listing 2: `StateExplorer.step` (simplified).

```
def step(self) -> Optional[SemanticState]:
    if not self.frontier:
        return None
    state = self._strategy_pop()        # BFS: deque.popleft(); DFS:
        stack.pop()
    for move in self.space.enabled_moves(state):
        succ = state.apply_transition(move)
        if succ.state_id not in self.visited:
            self.visited.add(succ.state_id)
            self._strategy_push(succ)
    self.local_constructor.consume(state)
    return state
```

The visited set is a Python `set` of state-id strings, ensuring $O(1)$ membership tests. Each unique state is therefore consumed by `LocalConstructor` exactly once, eliminating redundant section reconstruction.

2.3 Reachability and the finite-state assumption

Definition 2.4 (Finite-state program)**.** A program P is *finite-state* if the patch cover $\mathcal{P}$ is finite and each patch admits finitely many candidate sections. Equivalently, $|\mathrm{Reach}(\mathsf{SS})| < \infty$.

All programs that JuGeo currently handles are finite-state: the patch cover is determined by a finite CFG, and sections are drawn from a finite SMT-guided candidate pool. This assumption is made explicit in §7 and formalised in Lean 4.

3 Exploration Strategies

3.1 The `SearchStrategy` enum

The `SearchStrategy` enum in `generation/state_space/search_strategies.py` provides four values:

Variant	Frontier structure	Characteristics
`BFS`	FIFO queue	Complete; finds min-depth goal; $O(b^d)$ memory
`DFS`	LIFO stack	Complete iff acyclic; $O(bd)$ memory
`BMC`	FIFO + depth bound	Incomplete; bounded completeness guarantee
`HEUR`	Priority queue	Incomplete in general; guided by h

Here b is the branching factor (number of enabled moves per state) and d is the depth of the shallowest goal state.

3.2 Breadth-first exploration (Bfs)

Bfs maintains $\mathcal{F}$ as a FIFO queue. It guarantees that every state at depth k is processed before any state at depth $k+1$. Because the visited set prevents re-enqueuing, each state is dequeued at most once.

Proposition 3.1 (Bfs terminates on finite-state spaces)**.** *If* $|\text{Reach}(\mathsf{SS})| = N < \infty$*, then Bfs terminates after exactly* N *dequeue operations.*

Proof. By induction on d: at each depth level k, all states reachable in exactly k steps from σ_0 are distinct (by the hash-keying invariant) and each is enqueued exactly once. Since N is finite, the frontier eventually empties. □ □

3.3 Depth-first exploration (Dfs)

Dfs maintains $\mathcal{F}$ as a LIFO stack. In the presence of cycles (which can arise from Retract followed by Propose on the same patch), Dfs without a visited set would loop. With the visited set, Dfs also terminates on finite spaces (same argument as Bfs), but visits states in a different order that may bias toward deep, narrow paths.

Remark 3.2. Dfs is preferable when the first valid goal state is more important than the minimum-depth one. In practice, Dfs finds a complete local-section assignment faster than Bfs when the goal state is deep but reachable via a single path.

3.4 Bounded model checking (Bmc)

Bmc limits exploration to states reachable within depth d:

Definition 3.3 (Bmc strategy)**.** Given a depth bound $\mathsf{d} \in \mathbb{N}$, Bmc uses a FIFO queue augmented with depth annotations. A state σ at depth $k \geq \mathsf{d}$ is not expanded: its successors are not enqueued. If a goal state is reachable within depth d, Bmc finds it; otherwise it declares a *bounded non-termination*.

The key parameter d trades off coverage against time. Choosing $\mathsf{d} = \infty$ recovers Bfs.

Proposition 3.4 (Bmc partial completeness)**.** *Let* $p \in [0,1]$ *be the probability that a uniformly random successor of any state is a new patch coverage. Then Bmc with bound* d *covers a fraction*

$$f_{BMC}(\mathsf{d}) \;\geq\; 1 - (1-p)^{\mathsf{d}}$$

of the total coordinate patches in expectation.

Proof. Each exploration step covers a new patch with probability at least p. After d steps, the probability that a particular patch remains uncovered is at most $(1-p)^{\mathsf{d}}$. Summing over all patches and using linearity of expectation gives the bound. □ □

3.5 Heuristic search (Heur)

HEUR replaces the FIFO or LIFO discipline with a priority queue ordered by a *heuristic function* $h\colon \Sigma \to \mathbb{R}$. The default heuristic in JUGEO is the *coverage fraction*:

$$h(\sigma) \;=\; -\frac{|\mathrm{dom}(\alpha)|}{|\mathcal{P}|}$$

(negated because Python's `heapq` is a min-heap). States with more assigned patches are explored first. When h is an admissible heuristic (i.e. never overestimates the remaining cost), HEUR reduces to A* and is optimal.

Listing 3: Priority-queue pop for HEUR.

```python
class BestFirstStrategy(SearchStrategy):
    def pop(self, frontier: list) -> SemanticState:
        _, state = heapq.heappop(frontier)
        return state

    def push(self, frontier: list, state: SemanticState) -> None:
        priority = -len(state.assignments) / max(1, self.num_patches)
        heapq.heappush(frontier, (priority, state))
```

4 Local Construction

4.1 `LocalConstructor`

The `LocalConstructor` class (in `generation/local_construction/`) accepts a semantic state σ and produces a *local section* ℓ_σ at the coordinate encoded in $\sigma.\mathit{assignments}$.

Definition 4.1 (Local section from state)**.** Given a semantic state $\sigma = (\alpha, \Omega, \mathcal{T})$ and a coordinate $\mathsf{c} \in \mathrm{dom}(\alpha)$, the *local section at* c *induced by* σ is the restriction

$$\ell^\sigma_{\mathsf{c}} \;=\; \alpha(\mathsf{c})$$

together with the sub-bundle of Ω and $\mathcal{T}$ scoped to c.

The construction process uses three sub-algorithms:

1. **Candidate proposal** (`LoopStatus.RUNNING`): the constructor generates candidate sections using SMT-guided synthesis, type-directed enumeration, or LLM suggestion.
2. **Interface discipline check** (`InterfaceDiscipline`): the candidate is tested against the boundary constraints $\partial\mathsf{c}$ inherited from adjacent patches. Two sections are *compatible* iff their interface disciplines $D_u \sqcap D_v$ are jointly satisfiable.
3. **Acceptance** (`LoopStatus.SUCCEEDED`): if the candidate passes all discipline checks and closes the scoped obligations, it is recorded as ℓ^σ_{c} and the loop status transitions to `SUCCEEDED`.

Listing 4: `LocalConstructor.consume` (simplified).

```python
def consume(self, state: SemanticState) -> ConstructionStatus:
    for coord, section_id in state.assignments.items():
```

```
        loop = self._get_or_create_loop(coord)
        if loop.status == LoopStatus.SUCCEEDED:
            continue                              # already built
        candidate = self._synthesise(coord, section_id, state)
        if self._check_interface(candidate, coord):
            loop.accept(candidate)
            self._status[coord] = ConstructionStatus.COMPLETE
        else:
            loop.reject(candidate)
            self._status[coord] = ConstructionStatus.IN_PROGRESS
    return self._aggregate_status()
```

4.2 `ConstructionStatus` and state machine

Each coordinate's construction follows the status machine:

$$\texttt{PENDING} \xrightarrow{\text{init}} \texttt{IN\_PROGRESS} \xrightarrow{\text{accept}} \texttt{COMPLETE} \to \texttt{VERIFIED}$$

with an additional `FAILED` sink reachable from `IN_PROGRESS` when the budget is exhausted. The orchestrator tracks this status for all coordinates and reports aggregate progress to the explorer.

5 Construction Orchestration

5.1 Multi-coordinate management

When the state explorer visits a state σ, multiple coordinates in $\mathrm{dom}(\alpha)$ may be ready for local section construction. The `ConstructionOrchestrator` manages this concurrently:

Definition 5.1 (Orchestrator)**.** The *construction orchestrator* Orch is a function

$$\mathsf{Orch} \colon \Sigma \times \mathcal{L}^{<\omega} \longrightarrow \mathcal{L}^{<\omega} \times \mathsf{CS}$$

that takes the current state and the already-built sections, attempts construction for all new coordinates in $\mathrm{dom}(\alpha(\sigma)) \setminus \mathrm{dom}(\mathcal{L})$, and returns an updated section pool together with a status indicator.

The orchestrator ensures:

- **Idempotence**: constructing a section for a coordinate that already has one in $\mathcal{L}$ is a no-op.
- **Monotonicity**: $|\mathcal{L}|$ is non-decreasing across all orchestrator calls.
- **Interface consistency**: newly added sections are checked for compatibility with all existing adjacent sections.

5.2 Coordination graph

The orchestrator maintains a *coordination graph* $G = (V, E)$ where $V = \mathcal{P}$ and $(u, v) \in E$ iff patch u directly depends on patch v (e.g. a function call, a shared variable, or a treaty boundary). When a new section is accepted at u, all neighbours $v \in N_G(u)$ are notified so they can update their interface discipline D_v.

Listing 5: Orchestrator dispatch (simplified).

```
class ConstructionOrchestrator:
    def dispatch(self, state: SemanticState) -> ConstructionStatus:
        new_coords = (
            set(state.assignments) - set(self.built_sections)
        )
        for coord in self._topological_order(new_coords):
            result = self.constructor.consume_coord(coord, state)
            if result == ConstructionStatus.COMPLETE:
                self.built_sections[coord] = result
                self._notify_neighbours(coord)
        return self._aggregate()
```

5.3 Parallelism

The orchestrator supports two parallelism modes:

1. **Sequential**: coordinates are constructed in a topological order of G. This ensures that interface discipline constraints from upstream patches are available before downstream patches are processed.
2. **Parallel with barrier**: independent coordinates (those with no shared edges in G) are constructed concurrently using Python `asyncio` tasks. A barrier synchronises at each topological level before proceeding to the next.

In the sequential mode, the orchestrator is a special case of a *coordinated elaboration* (Paper 14, §4): the coordination graph defines the elaboration order.

6 Completion Conditions

6.1 The `CompletionCondition` abstraction

Exploration terminates when a `CompletionCondition` is satisfied. The abstraction separates the *what* (completion criterion) from the *how* (search strategy):

Definition 6.1 (`CompletionCondition`)**.** A *completion condition* is a predicate

$$\mathsf{CC} \colon \mathcal{L}^{<\omega} \times \mathbb{N} \times \Sigma \longrightarrow \{\mathsf{true}, \mathsf{false}\}$$

taking the current section pool $\mathcal{L}$, the number of explored states k, and the current state σ. The explorer halts as soon as $\mathsf{CC}(\mathcal{L}, k, \sigma)$ is true.

The implementation provides four concrete conditions:

Condition	Description
`ALL_PATCHES`	All patches in $\mathcal{P}$ have an accepted section ($f = 1$).
`BUDGET`	The number of explored states k exceeds a fixed budget B.
`COVERAGE_FRAC`	$f \geq \theta$ for a configurable threshold θ.
`FRONTIER_EMPTY`	The frontier $\mathcal{F}$ is empty (full exhaustion).

Conditions can be combined conjunctively (`AND`) or disjunctively (`OR`) to express policies such as "stop when all patches are covered *or* the budget is exceeded."

6.2 Interaction with strategies

Each strategy pair with a natural default condition:

- BFS: `FRONTIER_EMPTY` (full reachability) or `ALL_PATCHES` (early exit on coverage).
- DFS: `ALL_PATCHES` (stop at first goal state).
- BMC: `BUDGET` with $B = |\text{Reach}(\mathsf{SS}, \mathsf{d})|$.
- HEUR: `COVERAGE_FRAC` with $\theta = 0.95$.

6.3 Completion certificate

When CC fires, the orchestrator issues a *completion certificate*: a record containing the final section pool $\mathcal{L}$, the coverage fraction f, the number of explored states, and the elapsed time. This certificate is stored in the judgment 8-tuple as part of the evidence bundle $\mathcal{E}$.

7 Bounded Completeness Theorem

We now state and prove the main theoretical result.

Theorem 7.1 (Bounded Completeness)**.** *Let P be a finite-state program with reachable state space* $\text{Reach}(\mathsf{SS}) = \{\sigma_0, \ldots, \sigma_{N-1}\}$ *and patch cover $\mathcal{P}$ of size n. Let $\mathcal{L}^*$ be the union of all local sections obtainable by any sequence of semantic moves from σ_0. Then* BFS *with completion condition* `FRONTIER_EMPTY` *produces a section pool $\mathcal{L}$ such that:*

(i) $\mathcal{L} = \mathcal{L}^*$ *(completeness),*

(ii) Each state in $\text{Reach}(\mathsf{SS})$ *is visited exactly once (no redundancy),*

(iii) The total number of steps is $O(N + |\delta|)$ (time proportional to the reachable state space).

Proof. We prove each part in turn.

(i) Completeness. We show that $\mathcal{L} \supseteq \mathcal{L}^*$. Let $\ell \in \mathcal{L}^*$ be any section constructible from σ_0. Then there exists a path $\sigma_0 \xrightarrow{m_1} \sigma_1 \xrightarrow{m_2} \cdots \xrightarrow{m_k} \sigma_k$ with $\ell \in \text{sections}(\sigma_k)$. By induction on k: the base case $k = 0$ holds since σ_0 is enqueued initially. For the inductive step, assume σ_{k-1} is enqueued. When σ_{k-1} is dequeued, BFS generates all successors, including σ_k. Since $\sigma_k \notin \mathcal{V}$ (it has not been reached by any shorter path; if it had, its section would already be in $\mathcal{L}$), it is enqueued and eventually dequeued. When σ_k is processed, LocalConstructor adds ℓ to $\mathcal{L}$. Hence $\mathcal{L} \supseteq \mathcal{L}^*$.

The reverse inclusion $\mathcal{L} \subseteq \mathcal{L}^*$ is immediate: `LocalConstructor` only adds sections reachable from σ_0.

(ii) No redundancy. Each state $\sigma \in \text{Reach}(\mathsf{SS})$ is added to $\mathcal{V}$ at most once (when first discovered). A state in $\mathcal{V}$ is never re-enqueued. Therefore each state is dequeued at most once. Since every reachable state is eventually enqueued (by part (i)), it is dequeued exactly once.

(iii) Time complexity. Each state is dequeued once (N dequeue operations). Each dequeue generates the successors of σ: the number of successors is bounded by the out-degree of σ in δ. Summing over all states, the total number of successor-generation operations is $|\delta|$. Each visited-set membership test and insertion is $O(1)$ (hash set). Therefore the total time is $O(N + |\delta|)$. □ □

Corollary 7.2 (BFS coverage)**.** *Under the conditions of Theorem 7.1, the coverage fraction after* Bfs *terminates satisfies* $f = 1$*.*

Proof. By part (i) of the theorem, $\mathcal{L} = \mathcal{L}^*$. Since P is finite-state and all patches admit at least one constructible section (by the finite-state assumption), $\mathrm{dom}(\mathcal{L}) = \mathcal{P}$, so $f = |\mathrm{dom}(\mathcal{L})|/|\mathcal{P}| = n/n = 1$. □ □

Corollary 7.3 (DFS completeness under acyclicity)**.** *If* δ *is acyclic (the state space is a DAG), then* Dfs *with the visited set also achieves* $f = 1$*.*

Proof. In a DAG, Dfs with a visited set visits every reachable node exactly once by the same argument as Bfs; the LIFO discipline does not affect reachability. □ □

Remark 7.4. When δ has cycles, Dfs without a visited set may loop indefinitely. The `StateExplorer` *always* maintains a visited set, so Corollary 7.3 applies to the implementation even when the underlying state space has cycles.

8 Experiments

8.1 Benchmark setup

We evaluate the three main strategies (Bfs, Dfs, Heur) on the 300-program benchmark suite used throughout the JuGeo series. Programs range from 10 to 500 lines of Python, covering sorting algorithms, data-structure operations, arithmetic routines, and simple protocol implementations. The benchmark includes the 100 specification programs, 100 equivalence pairs, and 100 bug programs from Papers 1–10.

For each program we measure:

- **Coverage fraction** f: fraction of patches with an accepted local section.
- **Explored states**: number of `SemanticState` objects processed.
- **Wall time**: end-to-end time for the explorer loop.
- **Round count**: number of orchestrator dispatch calls.

All experiments run on a single core of a 3.2 GHz machine with 16 GB RAM. The SMT backend is Z3 4.13.

8.2 Strategy comparison

Table 1: Exploration strategy comparison on 300 programs. "States" = mean explored states per program; "Time" = mean wall time; "f" = mean coverage fraction; "Rounds" = mean orchestrator dispatch rounds.

Strategy	States	Time (ms)	f	Rounds	Completeness
Bfs	482	14.2	1.00	18.3	Full
Dfs	482	12.8	1.00	33.1	Full
Bmc (`d` = 5)	148	4.7	0.91	11.2	Bounded
Heur	207	6.9	1.00	17.8	Full

Bfs and Dfs achieve $f = 1.00$ on all programs (as guaranteed by the theorem). Heur matches full coverage while exploring only $207/482 = 43\,\%$ of the states that Bfs requires. Bmc at depth 5 misses $9\,\%$ of patches on average; raising the bound to $\mathsf{d} = 10$ brings it to $f = 0.98$.

8.3 Round efficiency

Dfs requires 33.1 orchestrator rounds on average vs. Bfs's 18.3. This is because Dfs visits deep states early, triggering interface-discipline negotiation with ancestors before those ancestors are fully constructed. The orchestrator must defer and retry these negotiations, inflating the round count. Bfs processes ancestors before descendants, so negotiations succeed on the first attempt.

8.4 Scaling with state-space size

Program size (lines)	|Reach|	**BFS time (ms)**	**DFS time (ms)**
≤ 50	≤ 120	3.1	2.9
51–100	121–400	8.7	8.2
101–250	401–1500	24.3	21.4
251–500	1501–6000	89.5	77.3

Figure 1: BFS and DFS wall time vs. program size.

Times scale near-linearly with |Reach|, consistent with the $O(N + |\delta|)$ bound. The slight super-linear component at larger sizes is attributable to hash-collision overhead in the visited set as $|\mathcal{V}|$ grows beyond the L3 cache.

8.5 Completion condition sensitivity

We compare two completion conditions for Heur: `ALL_PATCHES` (stop when $f = 1$) vs. `COVERAGE_FRAC` with $\theta = 0.90$. On programs where the last $10\,\%$ of patches require many high-depth states, `COVERAGE_FRAC` at $\theta = 0.90$ reduces the state budget by $61\,\%$ with a coverage cost of only 0.10. For verification-generation applications where high but not perfect coverage is acceptable, this trade-off is favourable.

9 Conclusion

We have developed a principled state space exploration framework for exhaustive local section generation in JuGeo. The key contributions are:

1. A formal **state space model** that captures the generation process as a directed graph of semantic states, with a precise notion of reachability (§2).
2. A **pluggable strategy** design—Bfs, Dfs, Bmc, Heur—that separates the frontier discipline from the termination criterion (§3).
3. A **local construction integration** showing how each visited state feeds the `LocalConstructor` and `ConstructionOrchestrator` (§§4–5).

4. A **Bounded Completeness Theorem** establishing that BFS achieves $f = 1$ for finite-state programs in $O(N + |\delta|)$ time, with no redundant state reconstruction (§7).

The framework closes the coverage gap identified in Papers 1–30 and provides the theoretical foundation for exhaustive evidence generation in the JuGeo system.

Future work. Three directions are most promising: (a) *Incremental exploration*: when a program changes, re-explore only the affected sub-graph of the state space. (b) *Compositional state spaces*: compose state spaces for sub-programs using the operadic machinery of Paper 14. (c) *Probabilistic strategies*: assign weights to transitions derived from coverage likelihood, enabling a principled *probabilistic completeness* result analogous to Proposition 3.4 but adaptive.

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